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Intro to Business Analytics and AI
Homework 3
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Association Rules and Clustering

Task 1: Association Rules

Screenshot of Rules Generated from SPSS Modeler

No.	Premises	Conclusion	Support	Confidence	LaPlace	Gain	p-s	Lift	Convic...
7	ArtBks	ChildBks	0.107	0.469	0.901	-0.350	0.016	1.179	1.134
8	ArtBks	CookBks	0.109	0.478	0.903	-0.348	0.015	1.155	1.123
9	CookBks, ChildBks	YouthBks	0.127	0.522	0.906	-0.360	0.069	2.180	1.591
10	DoItYBks	CookBks, ChildBks	0.132	0.524	0.904	-0.372	0.071	2.150	1.589
11	YouthBks	CookBks, ChildBks	0.127	0.531	0.909	-0.352	0.069	2.180	1.614
12	CookBks, ChildBks	DoItYBks	0.132	0.543	0.910	-0.355	0.071	2.150	1.634
13	GeogBks	ChildBks	0.141	0.574	0.916	-0.350	0.043	1.445	1.416
14	CookBks	ChildBks	0.244	0.589	0.880	-0.584	0.079	1.482	1.466
15	GeogBks	CookBks	0.147	0.601	0.921	-0.343	0.046	1.451	1.467
16	ChildBks	CookBks	0.244	0.613	0.890	-0.551	0.079	1.482	1.516
17	RefBks	ChildBks	0.115	0.617	0.940	-0.258	0.041	1.552	1.573
18	YouthBks	ChildBks	0.154	0.645	0.931	-0.325	0.059	1.622	1.696
19	DoItYBks	ChildBks	0.166	0.657	0.931	-0.339	0.066	1.654	1.758
20	YouthBks	CookBks	0.158	0.660	0.934	-0.321	0.059	1.594	1.722
21	RefBks	CookBks	0.127	0.678	0.949	-0.247	0.049	1.639	1.821
22	DoItYBks	CookBks	0.173	0.686	0.937	-0.332	0.069	1.656	1.864
23	CookBks, GeogBks	ChildBks	0.109	0.738	0.966	-0.186	0.050	1.857	2.299
24	CookBks, DoItYBks	ChildBks	0.132	0.764	0.965	-0.214	0.063	1.924	2.558
25	ChildBks, GeogBks	CookBks	0.109	0.772	0.972	-0.173	0.050	1.864	2.566
26	ChildBks, DoItYBks	CookBks	0.132	0.797	0.971	-0.199	0.064	1.927	2.893
27	CookBks, YouthBks	ChildBks	0.127	0.805	0.973	-0.189	0.064	2.027	3.097
28	ChildBks, YouthBks	CookBks	0.127	0.824	0.976	-0.182	0.063	1.991	3.332

0.5743440233236151

ChildBks

1. **What is the rule with the highest confidence? Most lift?**
2. **Please comment on the usefulness of above identified rule.**

Confidence tells us how prevalent the antecedent to the consequent transactions is compared to the antecedent. In this data set, the highest confidence or probability that an outcome would occur is if a person has read both child and youth books, then they are the most likely to read cookbooks.

Lift tells us the relationship of the two items in the dataset in terms of confidence. In this dataset, the strongest relationship is between people who have read both child books and youth books and those who read cookbooks.

If a person has read child books and youth books, then there is a 82.4% chance that they will read cookbooks. Similarly, the rule with the most lift (or correlation) is between those who have read both child books and youth books with cooking books, which is calculated as 3.332. When lift>1, this is indicatory that this rule is more useful than just selecting categories randomly.

Task 2: Perform a Cluster Analysis

Purchase Behavior

Screenshot of Cluster Profiles:

Cluster Model

```
Cluster 0: 237 items
Cluster 1: 64 items
Cluster 2: 301 items
Total number of items: 602
```

Avg. within centroid distance

```
Avg. within centroid distance: -12652247.707
```

Basis of Purchase

Screenshot of Cluster Profile:

Cluster Model

```
Cluster 0: 128 items
Cluster 1: 146 items
Cluster 2: 328 items
Total number of items: 602
```

Avg. within centroid distance

```
Avg. within centroid distance: -0.106
```

Screenshot of Joined Cluster Profile:

Cluster Model

```
Cluster 0: 320 items
Cluster 1: 127 items
Cluster 2: 155 items
Total number of items: 602
```

Open in Turbo Prep Auto Model Interactive Analysis

Row No.	cluster	average(AG...	average(ED...	average(Aff...	average(No...	average(Br...	average(Av...	count(clust...
1	cluster_0	3.230	4.292	17.796	3.931	16.708	11.533	318
2	cluster_1	3.094	2.811	10.291	3.016	10.559	7.563	127
3	cluster_2	3.277	4.542	20.942	3.542	18.045	15.954	155

1. Describe the characteristics of the clusters formed.

   There were 3 clusters formed when the combining the data sets.

   Cluster 0 has an average age of 3.230, an average education of 4.292, average affluence of 17.796, average number of brands owned to be 3.931, average brand runs to be 16.708, average price of each item to be 11.533.
   Overall, there were 318 data entries that aligned with the characteristics of this cluster.

   Cluster 1 has an average age: 3.094, an average education of 2.811, average affluence of 10.291, average number of brands owned to be 3.016, average brand runs to be 10.559, average price of each item to be 7.563.
   Overall, there were 127 data entries that aligned with the characteristics of this cluster.

   Cluster 2 has an average age: 3.277, an average education of 4.542, average affluence of 20.942, average number of brands owned to be 3.542, average brand runs to be 18.045, average price of each item to be 15.954.
   Overall, there were 155 data entries that aligned with the characteristics of this cluster.

2. Which cluster method gives you the best method and why?

   Based on the data analysis and deductive reasoning, I would say that cluster model 1 would give us the best method. The other models (cluster 0 and 2) have the average age of the consumer to be about 3 to 4 years old with an average educational time being around 4 years in school. Many children are not immediately put into school when they are born, so this doesn't make logical sense. Cluster 1, on the other hand estimates an approximate age of 3.094 and the average educational time to be 2.811. Even though this remains close (age almost equals time in school), this could be explained by daycare or nursery school education, thus making it more realistic than the other clusters.